Roll No.: .....

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech CSE (All specialization) Semester: IV

Name of the Course: Design and analysis of Algorithms Course Code: TCS 409

Time: 3 Hours **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**

ii. Answer any two sub questions among a, b & c in each main questions

iii. Total marks in each main question are 20.

iv. Each sub-questions carries 10 marks

Q1	(20Marks)	
(a)	Differentiate between worst case, best case, and average case time complexity with suitable examples.	CO1
(b)	Solve the recurrence relation T(n) = 2T(n/2) + n log n using recursion tree method.	CO2
(c)	Write a Python/C++/C/JAVA function to compute the nth Fibonacci number using dynamic programming. Discuss its time and space complexity.	CO2
Q2		
(a)	Explain the working of Heap Sort with a dry run on the array [4, 10, 3, 5, 1].	CO2
(b)	Implement the Quick Sort algorithm in any programming language. Explain the choice of pivot and its effect on performance.	CO2
(c)	Define internal and external sorting. When do we prefer external sorting? Give an example scenario.	CO2
Q3		
(a)	Explain how the Union-Find data structure can be used to detect cycles in an undirected graph.	CO3
(b)	Write pseudo-code for Floyd-Warshall algorithm and illustrate its working with an example graph.	CO5
(c)	Compare the Bellman-Ford and Dijkstra algorithms in terms of working, complexity, and applications.	CO5
Q4		
(a)	Discuss overlapping and optimal substructure properties in the context of Longest Common Subsequence (LCS). Solve LCS for "AGTABGTRE" and "GAGXTXAGYRBEFM".	CO2
(b)	Describe the Job Sequencing Problem with Deadlines. Provide an algorithm and demonstrate with an example.	CO2
(c)	Solve Matrix Chain Multiplication problem for matrices A1(10x20), A2(20x30), A3(30x40), and A4(40x30) using DP.	CO4
5		
(a)	What are the different collision handling techniques in hashing? Discuss their pros and cons.	CO5
(b)	Implement the Knuth-Morris-Pratt (KMP) string matching algorithm in C/C++/Python. Explain prefix function	CO6
(c)	Explain the Subset Sum problem. Why is it considered NP-Complete? Give an example instance and its brute-force solution	CO5